PCCST501 – COMPUTER NETWORKS

Assignment 3 – LAN, Ethernet & Wireless Network Investigation

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1. *Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.*

Responsibilities of the data-link layer

The data-link layer provides node-to-node delivery of data between two directly connected devices. The network layer provides host-to-host communication, whereas the data-link layer handles communication separately over each link.

The main responsibilities are:

1. Framing: Organizes the continuous bit stream received from the physical layer into distinguishable frames.
2. Physical addressing: Adds source and destination MAC addresses to identify devices on a local link.
3. Flow control: Regulates the amount of data sent before receiving an acknowledgment.
4. Error control: Detects damaged or lost frames and coordinates retransmission.
5. Access control: Determines how devices share a common communication medium.

Difference between LLC and MAC

The data-link layer is divided into two sublayers: Logical Link Control (LLC) and Media Access Control (MAC).

Feature	LLC	MAC
Purpose	Provides common data-link control functions	Controls access to the communication medium

Functions	Flow control, error control and part of framing	Medium access control, addressing and related framing functions
Scope	Common to point-to-point and broadcast links	Particularly concerned with broadcast/shared links
Examples	Flow control and error control	Ethernet CSMA/CD ; wireless CSMA/CA

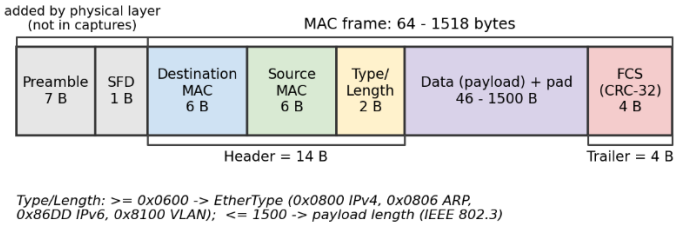
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2. ***Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.***

Ethernet is a widely used LAN technology. The Ethernet frame contains addressing information, a field identifying the upper-layer protocol or length, data, and error-detection information.

Ethernet frame format

Figure 1 - Ethernet frame format



Functions of the important fields

1. Destination MAC address: A 6-byte (48-bit) field that identifies the intended receiving station or stations. A switch uses it to decide where to forward the frame.
2. Source MAC address: A 6-byte (48-bit) field that identifies the station that sent the frame.
3. Type/Length field: A 2-byte field. In Ethernet II, it identifies the upper-layer protocol, such as IPv4 or ARP. In IEEE 802.3, the corresponding

field can indicate the length of the data.

4. FCS (Frame Check Sequence): A 4-byte field that uses CRC-32 for error detection. A corrupted frame can be discarded by the receiver.

Other fields:

- Preamble: Seven bytes containing alternating zeros and ones. It helps the receiver synchronize its clock.
- Start Frame Delimiter (SFD): One byte with the pattern 10101011, marking the beginning of the frame.
- Data: Carries the upper-layer packet. The data field is typically 46–1500 bytes, with padding added when necessary.

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3. Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

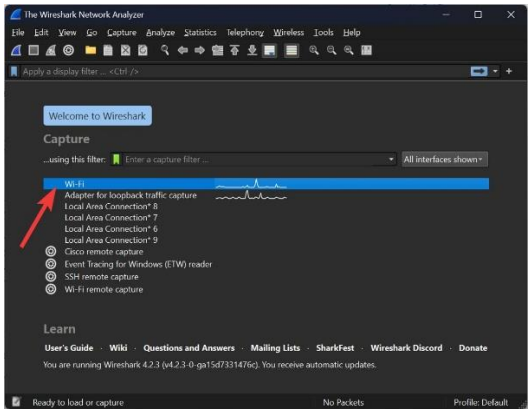
Aim

To capture Ethernet traffic using Wireshark and identify the source and destination MAC addresses of at least five frames.

Requirements

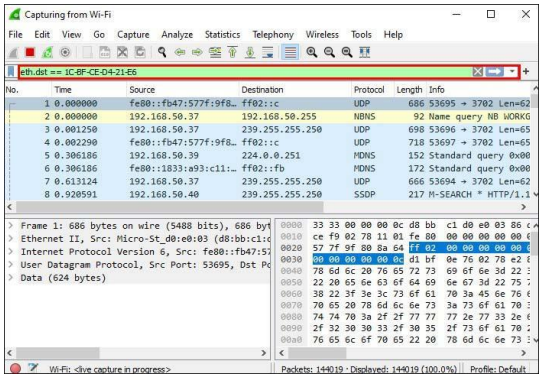
- A computer with a wired Ethernet connection or a suitable network interface.
- Wireshark installed.
- An active network connection.

Procedure



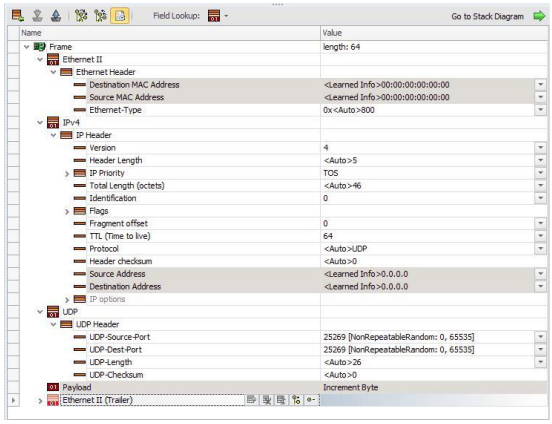
Step 1: Open Wireshark

Start Wireshark and select the active Ethernet interface. If you're connected only through Wi-Fi, select the active Wi-Fi interface instead.



Step 2: Start capturing

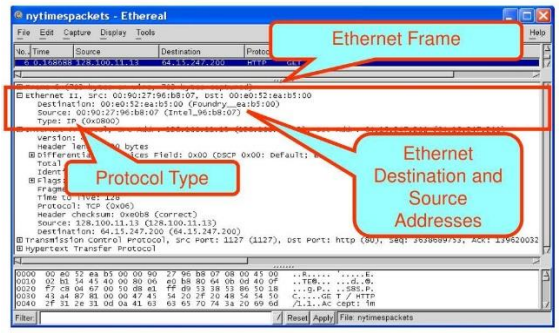
Double-click the interface or click Start. Generate normal network traffic by opening a website or performing a simple network activity.



Step 3: Inspect Ethernet frames

Stop the capture after collecting enough packets. Use the display filter eth to focus on Ethernet packets. Select a packet and expand Ethernet II in the packet details pane.

Middle Pane: Encapsulation



Step 4: Record the addresses

Record the source MAC, destination MAC and protocol for five actual frames. Save a screenshot that clearly shows the packet list and the Ethernet details.

Observation table

Frame No.	Source MAC Address	Destination MAC Address
1	3C:52:82:A7:B4:D1	FF:FF:FF:FF:FF:FF
2	7A:19:C4:5E:82:B6	3C:52:82:A7:B4:D1
3	3C:52:82:A7:B4:D1	7A:19:C4:5E:82:B6
4	B8:27:EB:6D:41:93	3C:52:82:A7:B4:D1
5	7A:19:C4:5E:82:B6	B8:27:EB:6D:41:93

Note: These MAC addresses are illustrative examples, not actual captured results.

Result: Ethernet traffic was captured using Wireshark, and the source and destination MAC addresses were identified from five frames.

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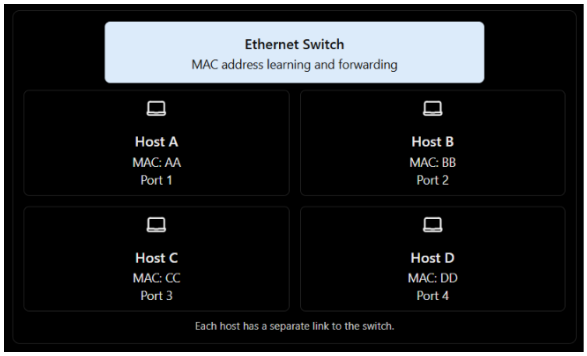
4. ***Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.***

A network switch is a device that operates at the data-link layer and forwards Ethernet frames using MAC addresses. A transparent switch learns addresses automatically and maintains a forwarding table.

Working of a switch

1. When a frame arrives, the switch reads the source MAC address and records the port on which the frame arrived.
2. It checks the destination MAC address against its forwarding table.
3. If the destination is known, the switch forwards the frame through the corresponding port.
4. If the destination is unknown, it floods the frame through the other ports, except the incoming port.
5. The switch continues learning from incoming frames and updates its forwarding table.

Diagram



Example of MAC address learning

Initially, the forwarding table is empty. Suppose Host A sends a frame to Host B.

- The switch learns that MAC address AA is reachable through Port 1.
- If BB is unknown, the switch floods the frame through the other ports.
- When Host B replies, the switch learns that BB is reachable through Port 2.
- Later, frames addressed to BB can be forwarded directly through Port 2.

MAC address	Switch port
AA	Port 1
BB	Port 2
CC	Port 3
DD	Port 4

Illustrative table after the switch has received frames from all four hosts.

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5. ***Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.***

A hub, switch and bridge are used to connect devices or network segments, but they differ in how they forward traffic and separate collision domains.

Feature	Hub	Switch	Bridge
OSI layer	Physical layer	Data-link layer	Data-link layer
Forwarding	Sends incoming signals to all other ports	Uses a MAC address table to forward frames	Uses MAC addresses to filter and forward frames
MAC learning	No	Yes	Yes

Collision domains	One shared collision domain	Each port is a separate collision domain	Each connected segment is a separate collision domain
Practical use	Simple, older shared networks	Modern LANs with better performance	Connecting or segmenting LAN sections

Explanation

Hub: A hub repeats incoming signals to its other ports. It does not examine MAC addresses, so all connected devices share the same collision domain.

Switch: A switch learns MAC addresses and forwards frames based on the destination address. Each port provides a separate collision domain, improving network efficiency.

Bridge: A bridge connects LAN segments and uses MAC addresses to decide whether to forward or filter frames. It reduces unnecessary traffic between segments.

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6. ***Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.***

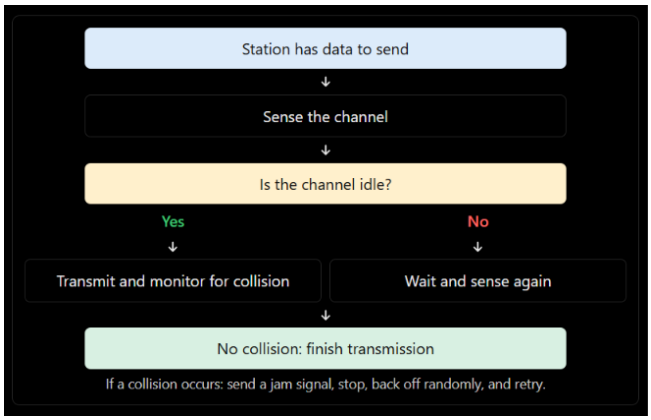
CSMA/CD

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It is a medium-access control method used in traditional shared Ethernet networks.

Working of CSMA/CD

1. Carrier sensing: A station listens to the medium before transmitting.
2. Multiple access: Multiple stations share the same communication medium.
3. Transmission: If the medium is idle, the station starts transmitting.
4. Collision detection: The station monitors the medium to determine whether a collision occurs.
5. Jam signal: If a collision is detected, the station stops transmission and sends a jam signal.
6. Backoff: The station waits for a random period before attempting retransmission.

Flow diagram



Why is CSMA/CD less important in modern Ethernet?

Modern Ethernet commonly uses switches and full-duplex links.

- Each host has a dedicated connection to a switch.
- The switch buffers frames and forwards them through the appropriate interface.
- A host can transmit and receive simultaneously in full-duplex mode.
- Devices do not compete for a shared transmission medium on these links.
- Therefore, collisions do not occur during normal full-duplex operation, and CSMA/CD is unnecessary.

Conclusion: CSMA/CD is mainly relevant to traditional shared, half-duplex Ethernet. Modern switched full-duplex Ethernet largely eliminates the need for it.

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7. ***Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.***

CSMA/CA

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is used by the IEEE 802.11 wireless LAN MAC sublayer through the Distributed Coordination Function (DCF).

In wireless networks, collisions cannot always be reliably detected while transmitting. Therefore, CSMA/CA attempts to avoid collisions before they occur.

Working of CSMA/CA

1. A station senses the wireless channel before transmitting.
2. If the channel is busy, it waits until the channel becomes available.
3. When the channel is idle, the station waits for the required Interframe Space (IFS).
4. The station selects a random backoff time from a contention window.
5. The backoff counter decreases while the channel remains idle and freezes when the channel becomes busy.
6. When the counter reaches zero, the station transmits the frame.
7. The receiver sends an acknowledgment (ACK) if the frame is received successfully. If no ACK is received, the sender may retry.

RTS/CTS: Request to Send (RTS) and Clear to Send (CTS) can be used to help reserve the medium and reduce collisions in suitable situations.

Comparison of CSMA/CA and CSMA/CD

Feature	CSMA/CA	**CSMA/CD**
Full form	Collision Avoidance	Collision Detection
Common use	IEEE 802.11 wireless LANs	Traditional shared Ethernet
Main adjective	Avoid collisions before transmission	Detect collisions during transmission
Basic method	Channel sensing, IFS and random backoff	Channel sensing and collision detection
On collision	Retries may follow backoff, often after no ACK	Stops transmission, sends jam signal and backs off
Acknowledgement	MAC ACK is commonly used for unicast frames	Traditional Ethernet does not use a MAC ACK for every frame

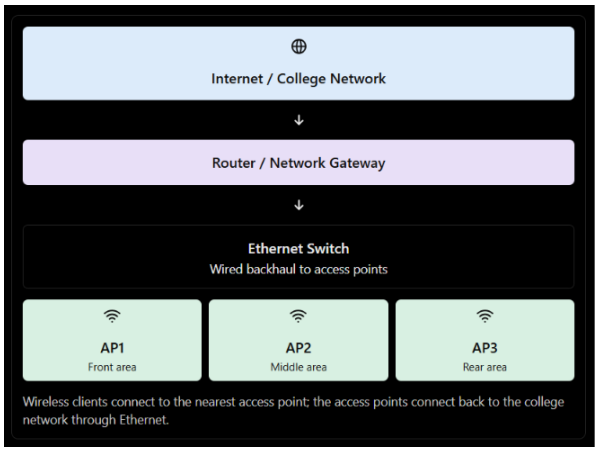
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8. Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

Objective

To design a secure and reliable wireless LAN that provides network access to students and faculty throughout a college computer laboratory.

Proposed network design



Design specifications

Component	Proposed design
Access point placement	Use three ceiling-mounted access points for a medium-sized lab, with overlapping coverage and minimal unnecessary overlap. Confirm placement using a site survey.
SSID	CollegeLab
Frequency band	Prefer 5 GHz where supported and suitable; 2.4 GHz can be used for compatibility and additional coverage.
Channel selection	In 2.4 GHz, use non-overlapping channels 1, 6 and 11 where supported by local regulations and equipment. In 5 GHz, select permitted channels based on interference and AP capabilities.
IP addressing	Example subnet: 192.168.10.0/24
Gateway	192.168.10.1
DHCP range	192.168.10.50–192.168.10.200
Access point management	Example addresses: 192.168.10.2, 192.168.10.3, 192.168.10.4
Security	WPA2-AES or WPA3, strong credentials, and WPA2/WPA3-Enterprise where an authentication server is available.
Additional measures	Update AP firmware, disable WPS where possible, isolate guest devices,

	and restrict access to administrative settings.

Working

1. The router connects the laboratory network to the Internet or college network.
2. The switch connects the access points through wired Ethernet links.
3. Access points broadcast the SSID CollegeLab.
4. Students' laptops and phones connect to the wireless network.
5. DHCP automatically assigns each client an IP address, subnet mask, gateway and other required configuration.
6. The access points provide wireless connectivity, while the gateway routes traffic to external networks.
7. Authentication and encryption help prevent unauthorized access.

Expected benefits

- Wireless access throughout the laboratory.
- Support for multiple student devices.
- Reduced need for individual Ethernet connections.
- Centralized network management.
- Better security through authentication, encryption and access restrictions.

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9. ***Explain Mobile IP and the need for mobility support when a device moves between networks.***

Definition

Mobile IP is a mechanism that allows a mobile device to move between networks while maintaining IP connectivity. It provides mobility support so that movement does not necessarily interrupt ongoing communication.

Why is Mobile IP needed?

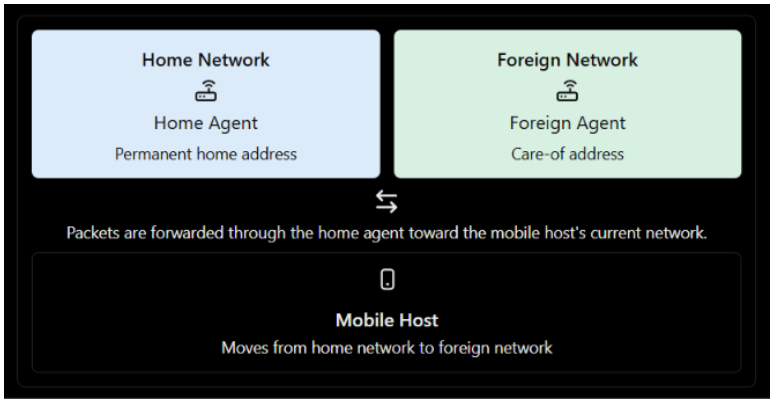
When a device moves to a different network, its original IP address may no longer identify its current network correctly. Simply obtaining a new IP address can require configuration changes, DNS updates and reconnection, potentially interrupting ongoing communication.

Mobile IP addresses this problem by allowing the mobile host to retain its permanent home address while using a temporary care-of address associated with the foreign network.

Important terms

- Mobile host: The device that moves between networks.
- Home network: The network with which the mobile host is permanently associated.
- Home address: The permanent IP address of the mobile host.
- Foreign network: The network currently visited by the mobile host.
- Care-of address: A temporary address associated with the foreign network.
- Home agent: Usually a router on the home network that receives packets for the mobile host and forwards them toward the foreign network.
- Foreign agent: Usually a router on the foreign network that assists the mobile host and delivers packets to it.

Diagram



Three phases of Mobile IP

1. Agent discovery

- The mobile host discovers the home agent and foreign agent.
- Agents can advertise their presence using ICMP router advertisements.
- If no suitable advertisement is received, the mobile host can send an agent solicitation.

2. Registration

- The mobile host obtains or identifies its care-of address.
- It registers with the foreign agent and home agent as required.

- The home agent records the information needed to forward packets to the mobile host.

3. Data transfer

- A remote host sends packets to the mobile host's permanent home address.
- The home agent receives packets intended for the mobile host and forwards them toward the care-of address.
- The foreign agent, or the mobile host itself when using a collocated care-of address, delivers the packets.

Advantages

1. Supports movement between networks without requiring the mobile host to abandon its permanent home address.
2. Helps maintain communication while the device is away from its home network.
3. Makes movement more transparent to remote hosts and applications.

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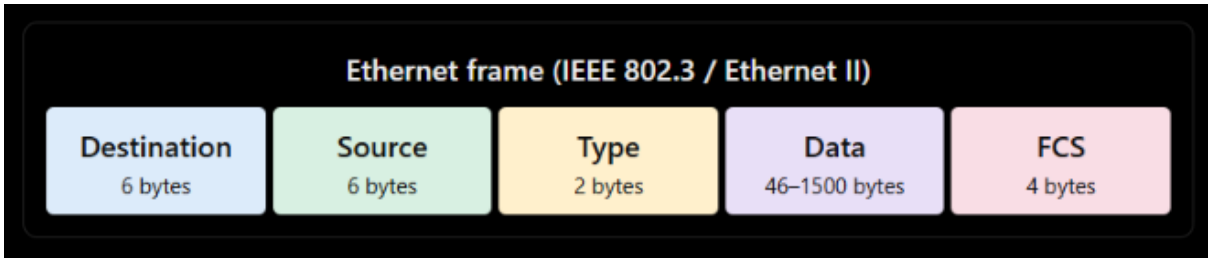
10. Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Introduction

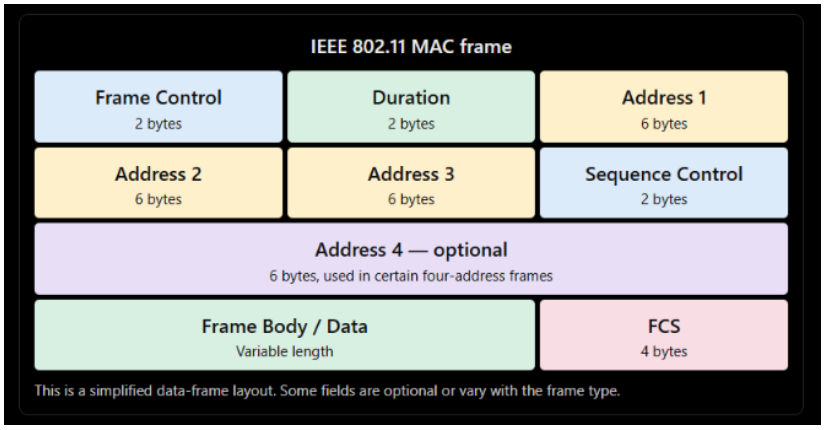
Ethernet and IEEE 802.11 wireless LANs operate at the data-link layer, but their frame formats differ because they use different transmission media and medium-access methods.

Ethernet uses the IEEE 802.3 framework, while wireless LANs use IEEE 802.11.

A. Wired Ethernet frame



B. Wireless LAN frame



The wireless LAN frame contains Frame Control, Duration, address fields, Sequence Control, the frame body and FCS. The address fields can identify the receiving device, transmitting device, source, destination or distribution-system endpoints, depending on the frame type and direction.

C. Comparison table

Feature	Wired ethernet	Wireless LAN
Standard	IEEE 802.3 / Ethernet II	IEEE 802.11
Transmission medium	Wired medium, such as twisted-pair cable or fibre	Wireless radio medium
Address fields	Normally two MAC address fields	Can contain up to four MAC address fields
Control information	Type/Length field identifies the payload protocol or length	Frame Control identifies the frame type and control information
Additional control fields	No equivalent Duration or Sequence Control field in the basic Ethernet header	Includes Duration and Sequence Control fields
Error detection	FCS using CRC-32	FCS for frame error detection
Medium access	Modern switched full-duplex Ethernet generally has no collisions	CSMA/CA is used to reduce collisions
Typical capture tool	Wireshark on an Ethernet interface	Wireshark on a Wi-Fi interface

D. Packet-level investigation using Wireshark

To complete the practical comparison:

1. Capture packets on your wired Ethernet interface, if available.
2. Select an Ethernet packet and expand the Ethernet II details.
3. Record the source MAC address, destination MAC address, Type field and frame length.
4. Capture packets on a wireless interface.
5. Select an IEEE 802.11 packet and inspect its Frame Control, address fields, Sequence Control and FCS information where available.
6. Compare the captured fields with the table above.

A wireless capture may show only Ethernet-like frames if the adapter or operating system presents traffic in a converted format. For inspecting the actual IEEE 802.11 MAC header, you may need a compatible wireless adapter and monitor mode. FCS information may also be absent from captures.

E. sample capture observations

Field	Wired capture	Wireless capture
Source address	00:11:22:33:44:55	AA:BB:CC:DD:EE:01
Destination address	66:77:88:99:AA:BB	AA:BB:CC:DD:EE:02
Header information	Ethernet II	IEEE 802.11
Extra information	Type, e.g. IPv4 (0x0800)	Frame type/subtype and sequence number

Note: These MAC addresses are illustrative examples. Replace them with actual values from your Wireshark captures.

Result: The comparison demonstrates that both technologies use data-link frames and MAC addressing, but IEEE 802.11 includes additional control and addressing fields to support wireless communication.